

CSEE 3827: Fundamentals of Computer Systems, Spring 2026

Lecture 0

Prof. Dan Rubenstein (danr@cs.columbia.edu)

Today's Lecture: Course Logistics

- Just covering the structure of the class
 - Grading
 - HW & Office Hour Policy (P-credit)
 - TAs
 - Things to sign up for (course calendar, OH availability)
 - Very high-level overview of course
 - Book / pre-reqs

Disclaimer on Harassment

- Me and TAs must report any harrassment / claims of harassment to <http://eoaa.columbia.edu>

**Your To-Do list for 3827 for
this week**

To-Do

- Get on EdStem through Courseworks (all sections will be through Section I)
- Take a crack at HW #0 (available on Courseworks) in Files under HW
- **By Tues 1/27:** Provide info on exam conflicts:
 - Midterm Conflict: March 5, 6-9pm (with March 3 and 4 as backup if no rooms available March 5),
 - Final Exam Conflict (May 8, 6-9pm)
- **By Saturday 1/24:** Provide your schedule for P-credit office hour (<http://uribe.cs.columbia.edu/sched/table.php>)

Lecture: 2 sections

- Tu,Th Sections
 - 10:10-11:25 (sec 1), 11:40-12:55 (sec 2)
- You are enrolled in one section
- Will merge course across sections as much as possible.
- Merged:
 - Courseworks/EdStem
 - Office Hour scheduling
 - HW
 - Exams!
- Not Merged (officially):
 - Lecture times

Section Policy

- The registrar has you registered in a specific section
- It really doesn't matter what section you are in (with small exceptions described next slide). So you can
 - Attend any section's lecture (or go to more than 1)
 - The midterm and final are “centralized” and time/place offered will not depend on your enrolled section.

Section Policy Exceptions

- If lecture room is overflowing (doubtful), priority given to:
 - a) people who haven't already attended an earlier version of the lecture
 - b) people who can't attend a later version of the lecture
 - c) people "officially enrolled"
- Exams: since not during class time, I am assessing conflicts now (see the links and dates on p. 5 of these slides)
 - Acceptable excuse: conflicting class, religious conflict, etc.
 - Indicate specific conflicts (class call #, specific time constraint if not class-related)
 - Not acceptable: "leaving town early", "busy week", etc.

Communications

Ways to communicate

- Class (in lecture)
- Office hours
- By email
- On EdStem

Email: subject header

- To email Prof. Rubenstein + TAs: 3827-staff@googlegroups.com
- Emailing just Prof. Rubenstein
 - Please include “3827” in the subject header of your email!!!
- I batch course emails a few times a week
- I search on “3827” in the subject. If it’s not there, I will miss your email.
- Subject headers that I often miss:
 - “Fundies”, “Your class”, “Issue”, “Help”, “Question”, etc.

Email example

To: danr@cs.columbia.edu
Subject: Urgent Fundies issue
Date: April 21, 2022

Hi Prof. Rubenstein,

My cat died and I'm real depressed, and HW is due tomorrow. The TA said they wouldn't give me an extension without your ok. Can you please let me know? Thanks!

Email example

To: danr@cs.columbia.edu

Subject: Urgent Fundies issue

Date: April 21, 2022

Hi Prof. Rubenstein,

My cat died and I'm real depressed, and HW is due tomorrow. The TA said they wouldn't give me an extension without your ok. Can you please let me know? Thanks!

No 3827 in the subject!!!

Email example

To: danr@cs.columbia.edu
Subject: Urgent Fundies issue
Date: April 21, 2022

Hi Prof. Rubenstein,

My cat died and I'm real depressed, and HW is due tomorrow. The TA said they wouldn't give me an extension without your ok. Can you please let me know? Thanks!

To: xxx@xxxxx
Subject: Re: Urgent Fundies issue
Date: May 5, 2022

Hi xxx,

Sorry to hear about your fish. Having a fish myself, I can appreciate how one dying can make things very difficult, and such a request has merit. However, I'm sorry but I only read this email today: since "3827" wasn't part of the subject, I missed it when doing a pass over the email related to the class. Please remember to include "3827" in future correspondence related to the class.

DR

Courseworks/EdStem

- We will use Courseworks as little as possible. Only for
 - EdStem
 - Posting files (HW, solutions, previous midterms and finals, etc.)
 - Whatever the TAs use if for (Gradescope?)
- We will be using EdStem exclusively for:
 - Notification of postings (lecture notes, HW, solutions, etc.) on Courseworks
 - Online Q&A

Grading

“Raw” Grade

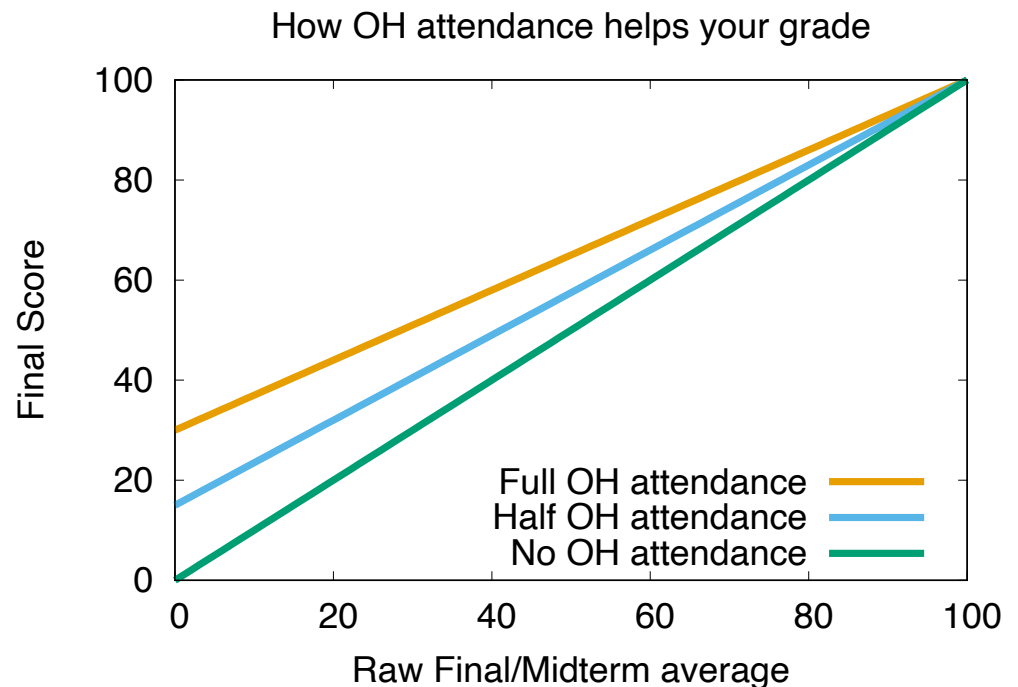
- “Raw” grade based on 3 parts:
 - **M**: midterm exam: midterm date Mar 5, 6:30-9:30pm (tentative)
 - **F**: end-of-term exam:
 - Trying to get Friday May 8, evening.
 - If not, probably May 14, (9-12am) section I official time
 - **H**: homework
- “Raw” score **S** = $10\% H + \text{Max}(30\% M + 60\% F, 45\% M + 45\% F)$
- If you have a **valid** conflict with these dates, let me (Prof. Rubenstein) know **before Jan 27 (Tuesday)** via the web forms on slide 5

“Boosted Grade”

- You can boost your final grade by attending office hours each week
- Office Hour participation score P : If you get a score of P for office hours (between 0 and 0.3), your raw score will be:
- $100 P + (1-P) S$ (recall $S = 10\% H + \text{Max}(30\% M + 60\% F, 45\% M + 45\% F)$)
- Let's see what this means...

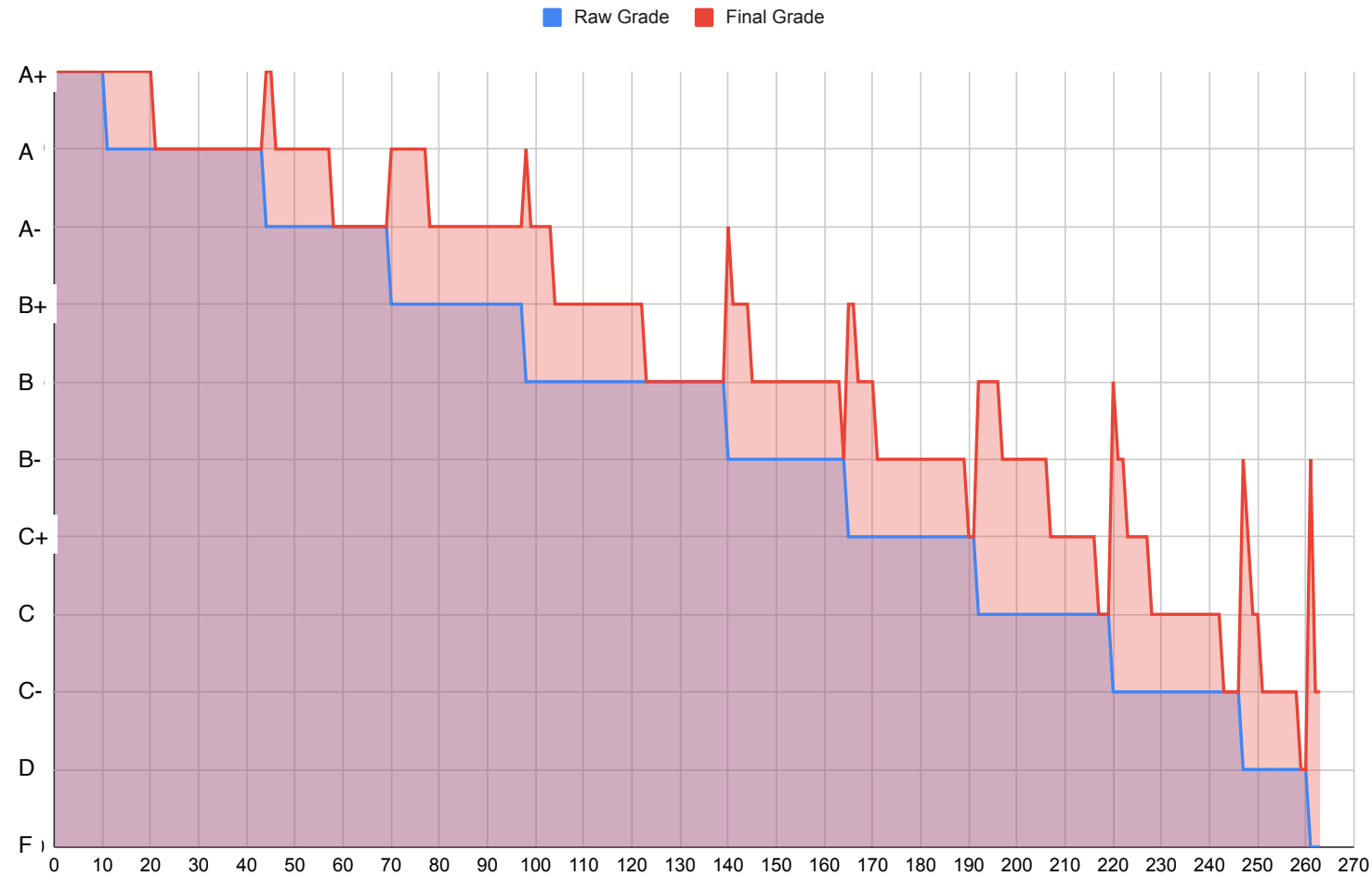
More on P

- Recall: Boosted score = $100P + (1-P) * S$
- $0 < P < 0.3$ (you still have to take midterm and final exams)
- Never go to OH: $P=0$, boosted score = S
- Always go: $P=.3$, boosted score = $30 + 0.7 * S$



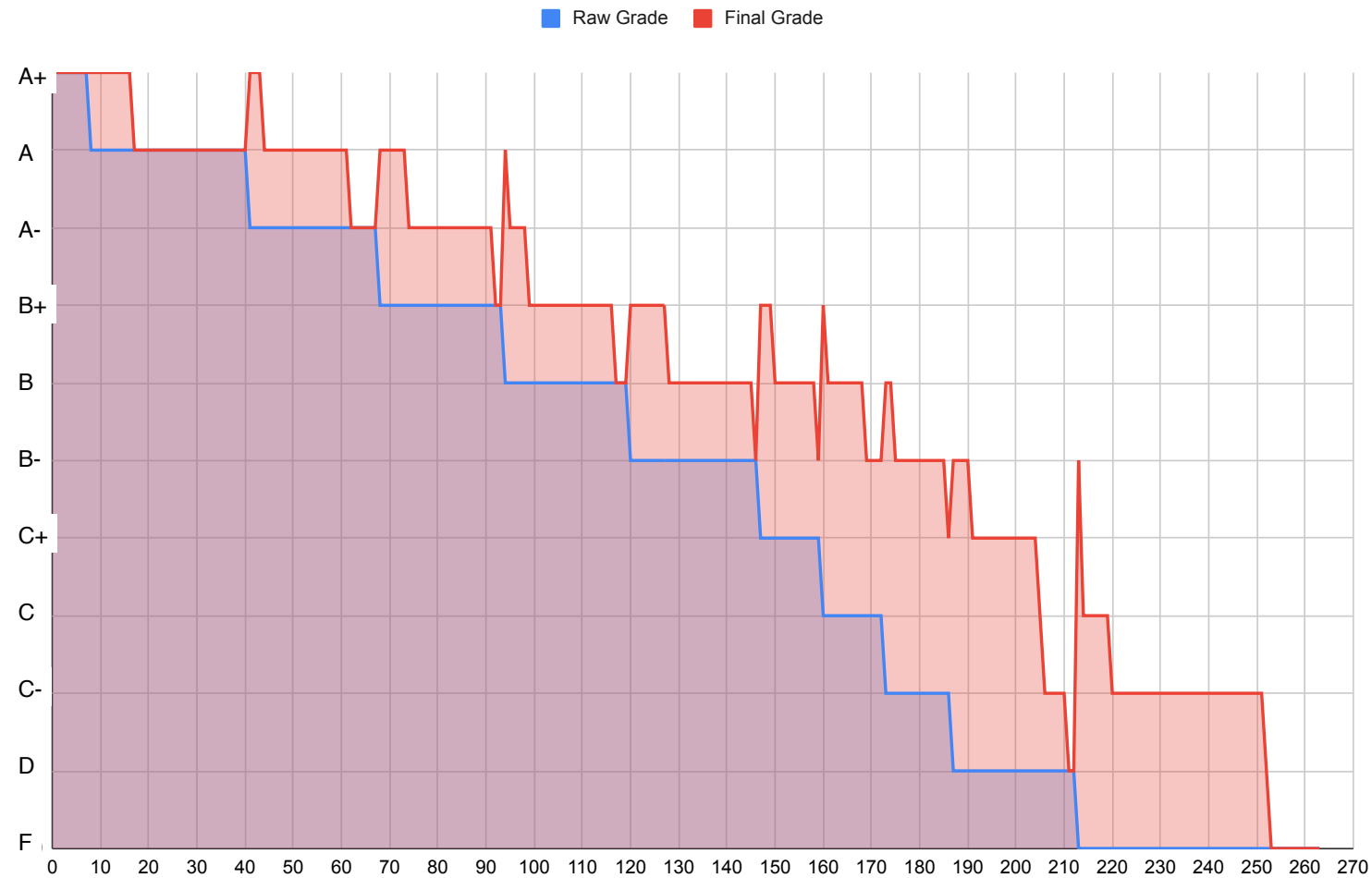
3827 S'24 grades View

3827 Final Grades Sp24



3827 S'23 grades View

3827 Final Grades Sp23



Homework

- Homework will be assigned (8 or 9 HWs)
- Homework will be discussed / worked on in Office Hours
- Homework **WILL BE submitted and graded on a 3 point scale:**
 - 0: didn't do it, 1: poorly done, 2: mostly right, 3: almost perfect
- Do it to learn / be prepared to talk about something in office hours
 - Doing the homework will prepare you for exams.

HW #0

- There is an initial homework (not due) to make sure you have requisite background for course
 - Modular arithmetic
 - basic logic
 - bit shifts
 - scientific notation
 - data flow
 - power-of-2 questions (ways to arrange k bits, gate depth)
 - coding / memory

Office Hours and P-Credit

Prof. Rubenstein's OH

- Weekly: W,Th 8am-9am
 - I might have to skip a day a week due to another commitment (check the OH calendar/EdStem for announcements)
- Or by appointment (email me with 3827 in the header):
 - it's best to put your availability in the email
 - It saves you having to wait for an email from me asking "what's your availability", and then going another round
 - PS - I have limited availability

Office Hour Score (P)

- Everyone will be assigned to a particular office hour
- You increase P by attending the full office hour
- You increase P by participating in the office hour
- You increase P by showing you've done your HW
 - Important: Enter your availability **by 1/24 11:59pm (Saturday night)** on <http://uribe.cs.columbia.edu/sched/table.php>
 - must provide at least 10 hour slots of potential availability
 - can select 1st, 2nd, 3rd choice per slot
 - choose hour blocks starting on any half-hour
- Algorithm will try to optimize all student requests

More on Office Hours

- You only get credit toward P by attending the office hour to which you were assigned
- You are still free to attend other office hours
- Prof. Rubenstein's OH not a P-credited OH, unless granted special permission
- TA's discretion to allow students to get credit for a non-assigned office hour (e.g., if can't make a session one week). Best if you ask in advance
- At most 1 credit per week (cannot make up for missed weeks by attending multiple OHs in a given week)
- Changing P-credit OH requires approval of TAs involved in change and Prof. Rubenstein

Office Hours Calendar

- Google/iCal calendar available listing our (Prof, TA) office hours
- Will be updated if office hours need to change in any week (e.g., cancelled, moved)
- Calendar link:
 - ical: <https://calendar.google.com/calendar/ical/16uc8kberc2b6dtltq6e49dv3k%40group.calendar.google.com/public/basic.ics>
 - html: https://calendar.google.com/calendar/embed?src=16uc8kberc2b6dtltq6e49dv3k%40group.calendar.google.com&ctz=America/New_York

Thoughts on Office Hours

- Offer a great opportunity to really learn/understand the material
- You should go even if it wasn't part of the grade requirement
- The TAs are all excited to meet/work with you

TAs

TAs

- There are 10 TAs:
- All TAs and Professor will be reachable via email address:
3827-staff@googlegroups.com
- also through EdStem

LLMs

- You won't have access to LLMs on the exams which is the bulk of your grade
- For HW, I can't stop you from using them
 - The only way to learn is by doing
 - So use them strategically for help if/when stuck

The Course

Why is this Course Important?

- **Basic CS Hardware Understanding:** how these machines we use every day work “under the hood”
- **Understanding of Hardware:**
 - Why is there a clock?
 - Apple’s new M5 chip has a 4.7 GHz clock: that’s 4,700,000,000 “instructions” per second). So what’s accomplished during an “instruction”? Why can’t the clock go faster?
 - What can the CPU really do?
 - We will learn basic CPU instruction set (MIPS) and build the architecture for it (from digital perspective)
 - We will cover how these instructions are combined to implement low-level language (e.g., C, Java) instructions

Why is this Course Important?

- **Understanding of Hardware (cont'd):**
 - What is memory (from a hardware perspective)? e.g., Why can M5 chip only support 16 Gig, 24 or 32 Gig of memory?
 - We will build memory (from a digital perspective) and learn some fundamentals about how it's used (storing data, storing addresses [pointers], implementing a stack)
- **New skill for your brain:** “Parallel thinking”: see <https://www.youtube.com/watch?v=8Pf8BkFLBRw> (warning: violence!)
- A computer’s “brain” works very differently than human brain
 - People think in sequence (one computation thread at a time): think & use
 - Circuits almost always built to process multiple “threads” at once, then combine (or throw away)

Why is this Course Important?

- Q: Which tech company is this?



Why is this Course Important?

- Q: Which tech company is this?



- A: NVIDIA

- They manufacture GPUs
- What's a GPU?
- In this course, we learn what a CPU is: big step toward understanding GPUs

Why is this Course Important? (Final Thoughts)

- **Useful** for future CS courses in obvious but also surprising places:
 - Basic programming: understanding various complexities of operations you are implementing: understanding why your code might be “slow”
 - Systems (OS/Networking/Databases): good design for these system attributes requires understanding “under the hood”
 - Programming Languages: they “compile down” or “interpret down” to the machine code “language” of the chip
 - ML/AI: part of their efficiency today is specialized chip design for AI systems. Previously, retooled GPUs (for graphics) to be used for AI.
 - Quantum computing (current quantum computing “programs” are very “circuit” based)
 - I’m sure there’s lots more...

ChatGPT #1

Q: Explain why a human should take a course in computer architecture

A: A course in computer architecture can provide a person with a deeper understanding of how computer systems work, including the design and organization of a computer's hardware and software components. This knowledge can be valuable for anyone working in fields related to computer science or engineering, such as software development, system administration, or network architecture. Additionally, taking a course in computer architecture can help a person to make more informed decisions about the design and implementation of computer systems, whether for personal or professional use. It can also provide a person with the ability to troubleshoot and optimize computer systems for better performance.

ChatGPT #2

Q: Why does a course in computer architecture help students with understanding in seemingly unrelated topics like AI, machine learning, user interface design, graphics, or quantum computing

A: A course in computer architecture can help students understand seemingly unrelated topics like AI, machine learning, user interface design, graphics, and quantum computing in several ways:

- Understanding hardware constraints: Understanding the basics of computer architecture can help students to better understand the constraints that hardware places on software and applications, including AI and machine learning algorithms, as well as quantum computing. This can help them design more efficient and effective systems.
- Understanding performance: Understanding the fundamentals of computer architecture can help students to optimize the performance of their systems, including AI and machine learning algorithms, as well as quantum computing. This can help them design systems that run faster and can process more data.
- Understanding parallel computing: Understanding the basics of computer architecture can help students to better understand how to take advantage of parallel computing, which is essential for many AI and machine learning algorithms, as well as quantum computing.
- Understanding user interface design: Understanding the basics of computer architecture can help students to understand how the hardware of the computer can affect the user experience, this can help them design more efficient and user-friendly interfaces.
- Understanding graphics: Understanding the basics of computer architecture can help students to understand how the hardware of the computer can affect the graphics processing and rendering, this can help them design more efficient and effective graphics algorithms.
- Understanding quantum computing: Understanding the basics of computer architecture can help students understand the underlying principles and limitations of quantum computing, which is a rapidly growing field that requires a deep understanding of computer architecture.

In summary, a course in computer architecture can help students with understanding in seemingly unrelated topics like AI, machine learning, user interface design, graphics, and quantum computing by giving them a deeper understanding of the underlying hardware constraints, performance optimization, parallel computing, user experience and graphics processing, and the underlying principles and limitations of quantum computing.

Course is about...

- Building a computer from a digital / logical perspective
 - Intro/Overview
 - working with booleans (1=TRUE, 0=FALSE, basic arithmetic, integer # representations)
 - use AND, OR, NOT operations
 - Build simple boolean expressions / circuits that are used in computing:
 - shifter, decoder, encoder, adder, etc.
 - learn how to simplify circuits (K-maps)
 - build basic circuits with “state” (latch, flip-flop)
 - build common stateful circuits (counter, register)

Course about (cont'd)

- Design stateful circuits via state machines
- RAM Memory design
- Register design
- Part 2 of course: use the circuits designed in part 1 to build a computer
 - MIPS Architecture and MIPS assembly
 - ALU design, control word processing
 - Building MIPS microprocessor (incl. branching and jumps)
 - Pipelined architecture
- Part 3: Some memory hierarchy details (caching, virtual memory)

Knowledge for class

- High school algebra (we'll do boolean instead of base 10), e.g., material covered in COMS 3203
- higher-level programming language (than assembly language), e.g., C-language, java, python
- Knowledge of assembly language very helpful, but not required (will be teaching MIPS)

Books (not required)

▼ Recommended Textbooks for purchase

	ISBN	Title	Authors	Publisher	Copyright Date	Price
1	9780133760637	Logic and computer design fundamentals	Mano, M. Morris,	Pearson,	[2016]	\$266.65
2	9780128201091	Computer organization and design : the hardwaresoftware interface	Patterson, David A.,	Morgan Kaufmann,	[2021]	\$99.95

- Will you really need them? Not in my opinion, but some students find them helpful

Course Lecture Slides

- Slides will be made available (from previous years) on Courseworks
- I will use slides this semester, with occasional writing on the board for clarification.
- After going on Courseworks, click on “Files”
- There are many, many, many slides
 - Lots of animations / visuals
 - They were created for lecture, not your “cram” guide

HW vs. Exams

- HW can be tedious and/or challenging: trying to accommodate students at all levels. Every question does have a purpose.
- Exam questions like medium-difficulty HW questions.
 - Exams meant to assess how well you know material. Not meant for everyone to get 100
 - To get 100, you must not only understand the material, but demonstrate aptitude for it.
 - We try to be generous with partial credit on exams
- **Do the HW:** it will help greatly with the exams
- **Attend Office Hours:**
 - lessens the “pain” from a bad test
 - we have better sense how well you understand the material

Lecture v. HW v. Exams

- Lecture explains the concepts
- HW is where you apply the concepts using what was taught in lecture
- Exams are where you demonstrate mastery of the concepts
- Some students don't see the connection between lecture, HW, exam: It's there!

A Learning Analogy

- **Lecture:** Teach how to take difference between two 3-digit numbers (e.g., 149-122, 381-226, 307-199)
 - You've got all the tools you need...
- **HW:** Take the difference between two 5-digit numbers (e.g., 30403 - 22785)
 - You're learning/showing how to take the tools and apply them beyond simple regurgitation
- **Exam:** Show how to compute 29468 - 31232? (Note: the larger value comes 2nd)
 - You're showing that you can think a bit “outside the box” and understand what's going on at a deeper, more fundamental level

Attendance / Announcements

- Attendance in lecture is not mandatory, but...
- Zoom: I'll provide recordings as long as I find the # of students in class sufficient, else recording stops...
 - Recording cannot be watched live (you'll sit in waiting room)
- Material is cumulative, the course builds on previous material: miss a lecture and the next one might be confusing
- It's your responsibility to find out what you missed
 - Don't ask Professor or TAs
 - Please limit such questions on EdStem (I won't answer)
 - Find a "class buddy" or 2 (now) - ask them.

Missed Exam

- You need a very good excuse (sickness with Dr note, family emergency, etc.)
- If you cannot take regular exam, I will give you a I-on-I oral exam.
 - It's scary (for you)
 - It's harder than the written exam, graded less leniently (no partial credit)
- Bottom Line: Don't miss the exam!

Final Thoughts

- Come to Lecture!
 - I've sat in classes: in person was always best
 - If you start falling behind, it's that much harder to catch back up
- Help your grade:
 - Go to office hours (don't be shy or ashamed): your tuition \$\$\$ pay for this, you should take advantage
 - Participate in Class

Final Final Thoughts

- **Don't cheat:**
 - it's wrong, selfish, unfair to other students, unfair to your own education
 - I will report suspicious activity to the Dean's office
 - I reserve the right to reduce a student's grade (all the way to an F if I feel it's warranted) for students caught cheating
- Be respectful when you email Professor or CAs

To-Do

- Make sure you can get on EdStem (through Courseworks)
- Take a crack at HW #0 (available on Courseworks) in “Files”
- Provide exam conflict times by 1/27 (see p.5 for link)
- When your schedule is finalized: provide availability for P-credit office hour (<http://uribe.cs.columbia.edu/sched/table.php>) by Saturday night, 1/24